

Hybrid Energy System Planning: Optimizing Solar and Wind Power for Maximum Efficiency and Cost Reduction

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Abstract— This project optimizes a hybrid renewable energy system with photovoltaic (PV) panels and wind turbines, aiming to minimize costs and maximize energy output. Advanced optimization techniques, including Simulated Annealing (SA), Genetic Algorithm (GA), Grey Wolf Optimization (GWO), and Artificial Bee Colony (ABC), are used to find near-optimal solutions within practical constraints. The decision variables include the number of PV panels and wind turbines, while the objective functions minimize costs and maximize energy output. Each algorithm offers unique advantages, with SA balancing global and local searches, GA emphasizing population diversity, GWO leveraging cooperative behavior, and ABC enhancing exploration and exploitation. Case studies reveal trade-offs between solar and wind configurations, demonstrating the potential of hybrid systems for cost-effective, energy-efficient solutions.

Keywords: Hybrid Energy System, Optimization Algorithms, Photovoltaic Panels, Wind Turbines, Metaheuristic Optimization

I. INTRODUCTION

The integration of renewable energy sources such as solar and wind has gained significant attention due to their sustainability and ability to mitigate environmental concerns. However, optimizing the design and operation of hybrid renewable energy systems is a challenging task because of the variability and intermittency of these resources. This necessitates the use of robust optimization techniques capable of handling complex, nonlinear, and multidimensional problems.

Metaheuristic algorithms, inspired by natural processes and phenomena, have emerged as powerful tools for addressing such optimization challenges. In this study, we investigate the performance of four metaheuristic optimization algorithms: Simulated Annealing (SA), Artificial Bee Colony (ABC), Genetic Algorithm (GA), and Grey Wolf Optimizer (GWO).

Each algorithm leverages a unique inspiration and strategy to explore and exploit the solution space effectively. SA which is suitable for finding near-optimal solutions in large search spaces. GA draws inspiration from the principles of natural selection and evolution. GWO is based on the social hierarchy and hunting strategies of grey wolves in nature. ABC emulates the foraging behavior of honeybee colonies.

By applying these algorithms to optimize a hybrid solar and wind energy system [1], we aim to evaluate their effectiveness in achieving cost reduction and energy output maximization. A comparative analysis of their results highlights the strengths

and weaknesses of each approach, providing valuable insights for optimizing renewable energy systems.

II. LITERATURE REVIEW

Numerous studies have investigated the optimization of hybrid renewable energy systems, focusing on the integration of photovoltaic (PV) panels and wind turbines to enhance efficiency and economic viability. These studies emphasize the importance of optimization techniques in balancing cost, energy production, and sustainability through the analysis of objective functions, decision variables, and system constraints.

A recent study (2024) highlighted two primary objectives in hybrid energy system optimization: minimizing the overall cost of energy (COE) and reducing the probability of power loss. By incorporating multiple renewable energy sources—including PV panels, wind and tidal turbines, hydraulic systems, biomass, and battery storage—the research achieved an economically optimal net cost of energy and power losses [2].

Another study explored distributed energy generation (DEG) systems, integrating economic, environmental, and social factors into optimization models. Artificial intelligence (AI)-based optimization methods have gained prominence, with Particle Swarm Optimization (PSO) identified as the most widely used approach, followed by genetic algorithm (GA) and harmony search algorithms (HS) [3]. Key findings include:

- **Algorithm Effectiveness:** Stochastic methods often yield higher net present costs (NPC), whereas deterministic methods generate higher-quality Pareto fronts for optimization.
- **Cost Reductions:** Cost reductions of 5% to 60% were observed under specific residential energy conditions.
- **Emission Reductions:** Emission reductions ranged from 45% to 90% through effective optimization strategies.

Bio-inspired optimization techniques, such as the Bee Algorithm, have been applied in hybrid renewable energy system optimization to enhance power efficiency [4]. Moreover, recent advancements in machine learning have enabled real-time optimization of energy generation and distribution [5]. These methodologies will be incorporated into our study to ensure optimal system performance.

Research on hybrid renewable energy systems integrating solar PV and wind energy highlights their growing importance

due to the limitations and environmental impacts of fossil fuels [6]. Key findings include:

- **Economic Feasibility:** Studies assess the economic viability of PV and wind systems using methodologies like the Loss of Power Supply Possibility (LPSP) technique and the Levelized Cost of Energy (LCE) method.
- **Optimization Techniques:** Various optimization methodologies, including graphical construction techniques, probabilistic methods, and AI-based algorithms, enhance the efficiency of hybrid systems.
- **Performance Assessment:** Simulation-based performance assessments use parameters such as solar radiation, wind speed, and load demand. Software tools like HOMER facilitate these evaluations.

Additionally, a bibliometric review by Cusa et al. [7] highlights the increasing global research interest in optimizing hybrid renewable energy systems for sustainable power generation. This study identifies key research trends, optimization techniques, and geographic areas suitable for hybrid energy implementation.

Recent research focused on optimizing the capacities of hybrid energy system components, such as PV panels and wind turbines, to maximize energy output while balancing economic objectives. The optimization process demonstrated the potential of hybrid renewable energy systems to achieve sustainable energy solutions through optimized decision variables [8].

A case study conducted in Antofagasta, Chile, examined the optimization of hybrid energy system components in a region with high solar irradiance and strong wind resources. The study, conducted at latitude $-22^{\circ}3'56.637''$ and longitude $-69^{\circ}22'19.2936''$ over an area of 2.47 km^2 , optimized the sizes of the PV panels and wind turbines to enhance energy output and minimize costs [9]. The findings from this study emphasize the significance of geographical and climatic factors in determining the feasibility of hybrid renewable energy systems, reinforcing the importance of site-specific optimization strategies.

Egypt offers a highly suitable environment for implementing hybrid renewable energy systems due to its abundant solar irradiance and favorable wind conditions [10]. Studies on solar radiation levels in Egypt confirm that the country receives some of the highest solar irradiance levels in the world, particularly in Upper Egypt [11].

To validate the proposed optimization approach, our research will focus on Zafrana [12], a coastal region in Egypt known for its high wind speeds and solar radiation. The region has been identified as a prime location for renewable energy projects due to its consistent wind resources and strong solar potential.

III. METHODOLOGY

A. Problem Formulation

1) Objective Functions

To address the problem, we first concentrated on an objective function aimed at minimizing the total cost of the

system within the given budget constraints. We will introduce the decision variables and constraints associated with this objective, which focuses on minimizing the total net cost.

$$\text{Cost}_{\text{total}} = X_s \cdot D_s + X_w \cdot D_w + D_m \quad (3.1)$$

where X_s represents the number of solar panels, D_s is the PV Panel cost, X_w is the wind turbine cost, D_w is the wind turbine cost and the D_m is the maintenance cost.

For the second objective function, our target was to maximize the energy output of wind turbines and photovoltaic panels. The main challenge lies in maximizing the energy output from our hybrid system in a specific layout while minimizing total costs.

$$E = \sum_{t=1}^n (\eta_s \cdot A_s \cdot G_t \cdot X_s + \frac{1}{2} \cdot \rho \cdot A_w \cdot v_t^3 \cdot C_p \cdot X_w) \Delta t \quad (3.2)$$

where η_s denotes the efficiency of the solar panels, A_s is the corresponding area of the solar panel, G_t represents the solar irradiance, ρ is the air density, v_t represents the wind speed, A_w is the swept area of the wind turbine and C_p is the power coefficient of the wind turbine.

2) Decision Variables:

- **Number of Photovoltaic Modules (X_s):** This represents the count of solar panels needed in the system.
- **Number of Wind Turbines (X_w):** The count of wind turbines included in the hybrid system
- **Type of the Pv panels:** From the standard Pv panels that affects its cost and area
- **Type of Wind turbines:** From the standard Wind turbines that affects its cost and area

3) Constraints:

- **Cost:** This involves installation, operation, and maintenance costs of the system components.
- **Distance Between Wind Turbines:** A minimum distance must be maintained between each pair of turbines to ensure optimal performance and prevent interference.
- **Total Area for Our System:**

$$A_t > X_s \cdot A_s + X_w \cdot [8 \times (6 + 2r)] \quad (1)$$

where r is the radius of each wind turbine. The total operational area accounts for the turbine's diameter plus a 6-meter margin of safety, multiplied by a length factor 8 to estimate the overall occupied space for each wind turbine.

For simplification, solar irradiance and wind speed were assumed to be constant during simulation runs. This assumption allows for isolating the effects of the optimization strategy itself. In real-world applications, solar and wind resources fluctuate dynamically. The use of constant values presents a limitation in accurately predicting system performance under varying conditions. Future studies will incorporate time-series environmental data

to improve the realism and resilience of the proposed models.

- **Standard Sizes for PV Panels and Turbines:**

- Standard Sizes for PV Panels Solar Areas: 1.89, 1.65, 1.9305, 1.95, 1.9602 m².
- Costs of PV Panels Sizes Respectively: 281, 245, 375, 475, 525 USD.
- Standard Sizes for Wind Turbines Swept Areas: 28, 32, 36, 40, 44, 48 m².
- Costs of Wind Turbine Sizes Respectively: 50k, 58k, 65k, 72k, 79.2k, 86.4k USD.

B. Simulated Annealing Algorithm

we apply the Simulated Annealing (SA) method, which is suitable for finding near-optimal solutions in large search spaces. The algorithm aims to explore the solution space while balancing between global exploration and local exploitation. Initially, we generated a starting configuration for the optimization problem, which included the number of solar panels (Xs), the number of wind turbines (Xw), and their available types. During the iterative process, we adjusted these parameters by changing the number of solar panels, the number of wind turbines, and the indices associated with both PV panels and wind turbines (one at a time, and the one we change is chosen randomly), allowing for an exploration of different configurations to find the optimal solution.

- Case Study 1: The algorithm attempts to reduce the total system cost by minimizing the objective function over 1000 iterations, 500 initial temp, 0.01 final temp.
- Case Study 2: The algorithm attempts to reduce the total system cost by minimizing the objective function by increasing the initial temp to 1000.
- Case Study 3: The algorithm attempts to reduce the total system cost by minimizing the objective function by decreasing the iterations to 500.

C. Genetic Algorithm

Another approach we took was the genetic algorithm for generating the optimal solution for our problem. Genetic algorithms solve optimization problems by mimicking the process of natural selection, where a population of candidate solutions evolves over generations through selection, crossover, and mutation, gradually improving and converging towards the optimal solution. our considerations are as follows, crossover: Whole arithmetic recombination, Mutation: adding noise, Parents: best.

- Case Study 1: In the first case study, the optimization process was conducted with a population size of 100 individuals. The arithmetic recombination in the crossover phase employed a weighting factor, α , set to 0.6, emphasizing the contribution of one parent over the other during offspring generation. This evolutionary algorithm was executed over 50 generations.

- Case Study 2: In the second case study, the optimization process was adjusted to run for the same iterations and populations but α was increased to 0.9
- Case Study 3: In the third case study, the population size was increased to 200, while maintaining the α value at 0.6 for the arithmetic recombination in the crossover phase and keeping the number of generations fixed at 50.

D. Grey Wolf Algorithm

Next, we tried the grey wolf algorithm for our problem which is a metaheuristic optimization technique inspired by the social hierarchy and hunting behavior of grey wolves in nature. It is designed to solve optimization problems by mimicking how grey wolves encircle and attack prey.

- Case Study 1: The algorithm was executed with a population size of 500 and 70 generations. This configuration serves as the baseline for evaluating the impact of parameter changes.
- Case Study 2: To assess the effect of population diversity, the population size was decreased to 100 while maintaining the same number of generations (70).
- Case Study 3: To evaluate the impact of reducing the search duration, the number of generations was reduced to 50 while keeping the population size fixed at 500.

E. Artificial Bee Algorithm

The Artificial Bee Colony (ABC) algorithm is a nature-inspired optimization technique modeled after the foraging behavior of honey bees. It mimics how bees collaboratively search for the most optimal food sources, representing potential solutions to optimization problems

- Case Study 1: The algorithm was executed with a population size of 50, 100 generations, limit of 10 and α is 0.3. This configuration serves as the baseline for evaluating the impact of parameter changes.
- Case Study 2: To assess the effect of population diversity, the population size was maintained at 50 while increasing the generations to 200. with limit 10.
- Case Study 3: To assess the effect of limit, the population size was maintained at 50 and the generations kept at 100. with limit increased to 20.

F. Machine Learning Algorithm

The concept revolves around optimizing renewable energy systems by combining data clustering and performance evaluation techniques. Using K-means clustering and PCA (Principal Component Analysis), the system groups feasible configurations of solar panels and wind turbines based on their energy efficiency and cost-effectiveness. This allows for identifying patterns and trade-offs between energy production and expenses.

The approach integrates a feasibility check to ensure solutions meet constraints like area and budget. An optimization metric evaluates configurations to find the most efficient and

cost-effective solution. Clustering provides insights into distinct groupings, helping stakeholders make informed decisions about resource allocation and energy system design. The visualization adds clarity by showing these trade-offs in a simplified 2D space, aiding strategic planning for sustainable energy solutions. Case 3 is the most effective configuration for this SA algorithm.

IV. RESULTS AND DISCUSSION

A. Simulated Annealing Algorithm

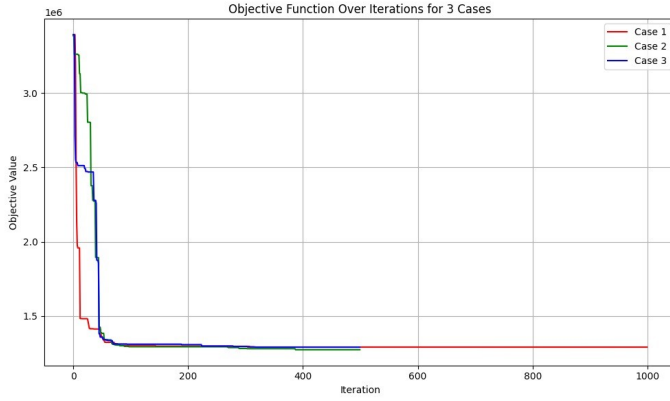


Fig. 1. Simulated Annealing Case studies

As shown in 1 Case 3 outperforms Cases 1 and 2, achieving the lowest objective function value early and maintaining it, indicating the best optimization efficiency. Case 2 performs better than Case 1 but stabilizes at a higher value, while Case 1 shows slower improvement, likely due to local minima. All cases converge around 100 iterations, emphasizing the need for proper parameter tuning to enhance exploration and avoid premature convergence.

B. Genetic Algorithm

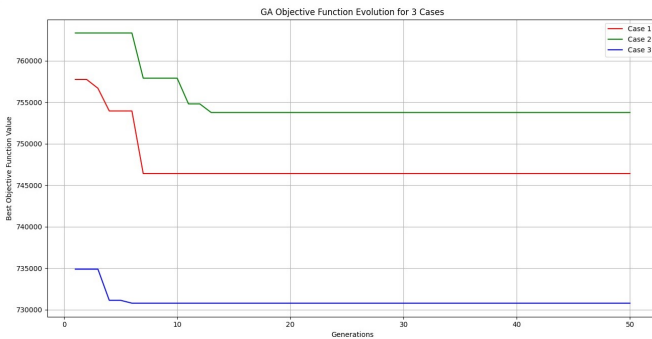


Fig. 2. Genetic Algorithm Case studies

As demonstrated in Figure 2 The genetic Algorithm (GA) demonstrates varying optimization performance across the

three cases. Case 3 achieves the lowest objective function value early, indicating superior optimization efficiency. In contrast, Cases 1 and 2 gradually improve but stabilize at higher values, suggesting potential local minima entrapment. All cases converge after around 20 generations, reflecting reduced exploration over time.

C. Grey Wolf Algorithm

The grey Wolf Optimizer (GWO) shows effective convergence in all cases. Case 3 and Case 2 achieve the lowest fitness values, indicating superior optimization. Case 1 starts higher, but drops early, stabilizing at a higher fitness. Case 2 improves late, suggesting sustained exploration, while Case 3 converges faster but with less exploration As shown in figure 3.

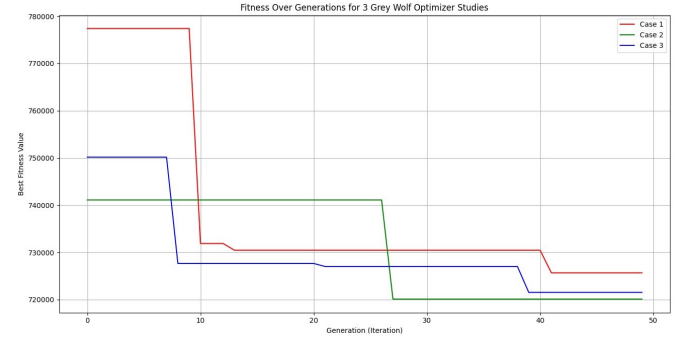


Fig. 3. Grey Wolf Case studies

D. Artificial Bee Algorithm

According to Figure 4 The Artificial Bee Colony algorithm shows a steady fitness improvement over iterations for all cases. Case 2 achieves the best final fitness, converging faster with sharper drops, indicating better exploration and exploitation. Case 3 follows closely, while Case 1 shows a slower but stable convergence. The results suggest that parameter tuning significantly affects the efficiency of ABC.

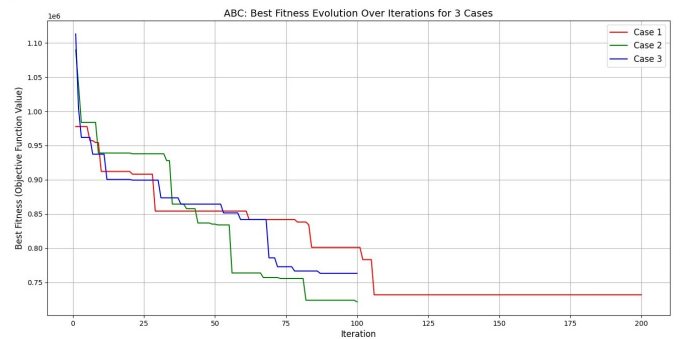


Fig. 4. Artificial Bee Case studies

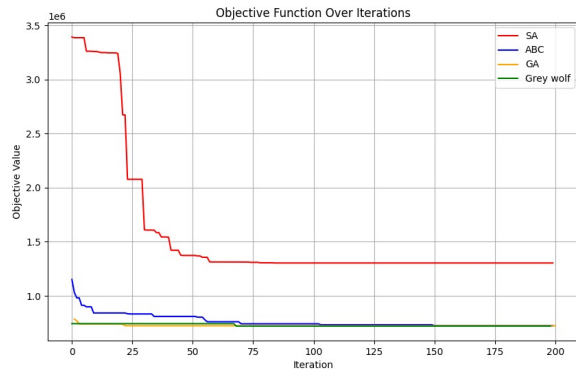


Fig. 5. Common Case Study

Figure 5 presents a common case study with a population size of 100 and 200 generations, demonstrating the convergence behavior of different optimization algorithms for the hybrid renewable energy system problem. The results indicate variations in convergence speed and final objective values, with some algorithms reaching near-optimal solutions significantly earlier than others. Notably, certain algorithms exhibit rapid initial improvement, while others show more gradual refinement over iterations. These results highlight the impact of algorithmic dynamics on optimization efficiency and solution quality in hybrid renewable energy system design.

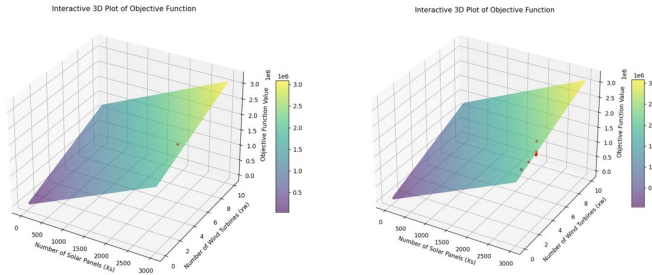


Fig. 6. Simulated annealing visualization

From Figure 6 The visualization showcases the performance of the Simulated Annealing (SA) algorithm in optimizing the hybrid energy system. The 3D plots represent the objective function values across varying numbers of solar panels (Xs) and wind turbines (Xw). The gradient color scheme indicates the objective function values, with darker regions representing lower costs. The red dots highlight the optimal solutions identified by SA, demonstrating its ability to converge to regions of higher efficiency. This visualization effectively illustrates how SA navigates the solution space to achieve optimal system configurations.

It can be observed in Figure 7 that this visualization shows the Genetic Algorithm's progression, starting with diverse solutions in the first generation and gradually converging

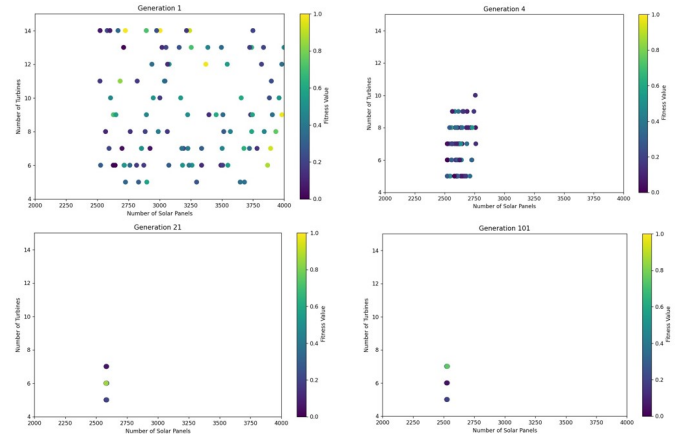


Fig. 7. Genetic Algorithm visualization

toward an optimal configuration by the 101st generation. Over time, the solutions focus on regions with higher fitness, demonstrating the algorithm's ability to refine and optimize effectively.

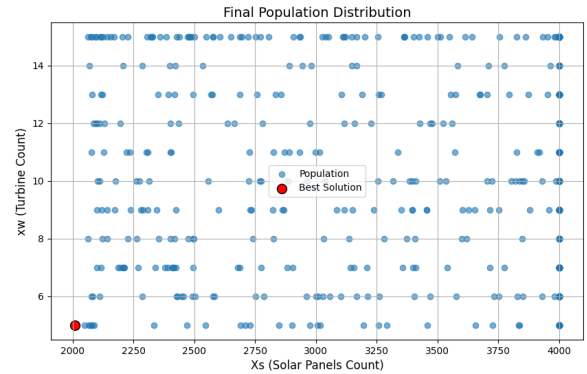


Fig. 8. Grey Wolf Optimizer

This visualization illustrates the final population distribution of the Grey Wolf Optimization algorithm. The blue dots represent the potential solutions in the population, where each point corresponds to a specific combination of solar panel count (Xs) and turbine count (Xw). The red dot highlights the best solution identified by the algorithm, which likely represents the optimal trade-off between the two variables to maximize the objective function. The spread of the blue dots indicates the diversity of solutions in the final generation, suggesting that the algorithm explored a wide search space while successfully converging on the best solution As depicted in Figure 8.

Figure 9 shows K-means clustering with PCA, highlighting three distinct clusters of feasible solutions based on energy production and cost. The centroids represent the central characteristics of each group, while PCA captures the key patterns

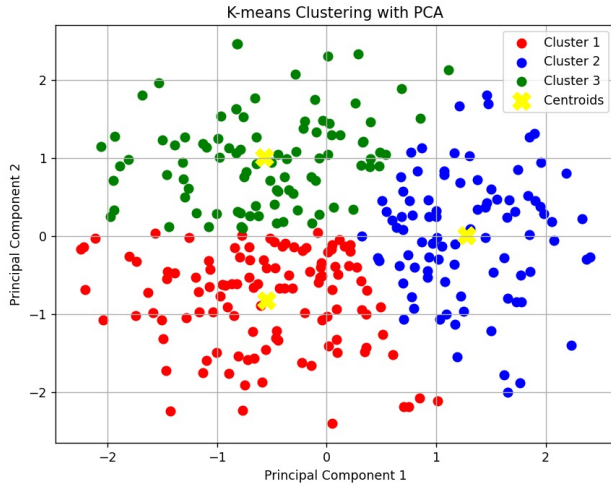


Fig. 9. K-Means Clustering with PCA

in reduced dimensions. The clusters may indicate different strategies (e.g., high-cost/high-energy vs. balanced trade-offs). Overlaps suggest some similarity between groups, and further analysis of cluster attributes can provide deeper insights into solution feasibility.

E. Performance Indices

	SA	GA	GW	ABC
Optimal Solution	[2835, 5, 0, 0]	[2149, 8, 1, 1]	[2000, 7, 0, 0]	[2075, 7, 0, 0]
Optimal Fitness Value	998575	855099	710500	785575
Mean Fitness	1265080	855099	721267.4	885464
Computational Time	0.0610 seconds	0.2249 seconds	0.04084 seconds	22 seconds

TABLE I
PERFORMANCE INDEX

As shown in the performance indices table, among the four optimization algorithms evaluated, Gray Wolf Optimizer (GWO) achieved the best fitness value with the fastest computational time, making it the most efficient choice. Genetic Algorithm (GA) performed well but required more time, while Simulated Annealing (SA) showed higher variability and stuck in local optima. Artificial Bee Colony (ABC), though effective in exploration, was the slowest, making it computationally expensive. Overall, GWO is the most optimal, GA balances quality and time and ABC is impractical for large-scale use without speed improvements as it demonstrated higher computational time compared to other algorithms. This is attributed to large number of solutions, extensive local search process, and slower convergence characteristics. To address this, future work could explore parameter tuning (e.g., colony size and number of iterations).

V. CONCLUSION AND FUTURE RECOMMENDATIONS

In this study, we investigated the optimization of hybrid energy systems composed of solar panels and wind turbines, aiming to achieve an optimal balance between energy production and system efficiency. The problem was formulated with constraints on resource allocation and energy demands, and was tackled using several metaheuristic algorithms. The results demonstrated the effectiveness of these algorithms in identifying feasible and near-optimal solutions within the design space.

Overall, the research contributes to the field of renewable energy optimization by providing a comparative framework and practical insights into metaheuristic algorithm performance. Future work could expand on this study by incorporating additional real-world constraints, such as environmental impact, dynamic energy demands, and system scalability, as well as testing more advanced optimization algorithms to enhance solution quality and robustness.

To further enhance the optimization process, we recommend expanding the solution array by incorporating additional system design variables to improve robustness and capture more complex relationships within the hybrid energy system.

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